

TM Forum Component

Bill Calculation

TMFC031

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Notice

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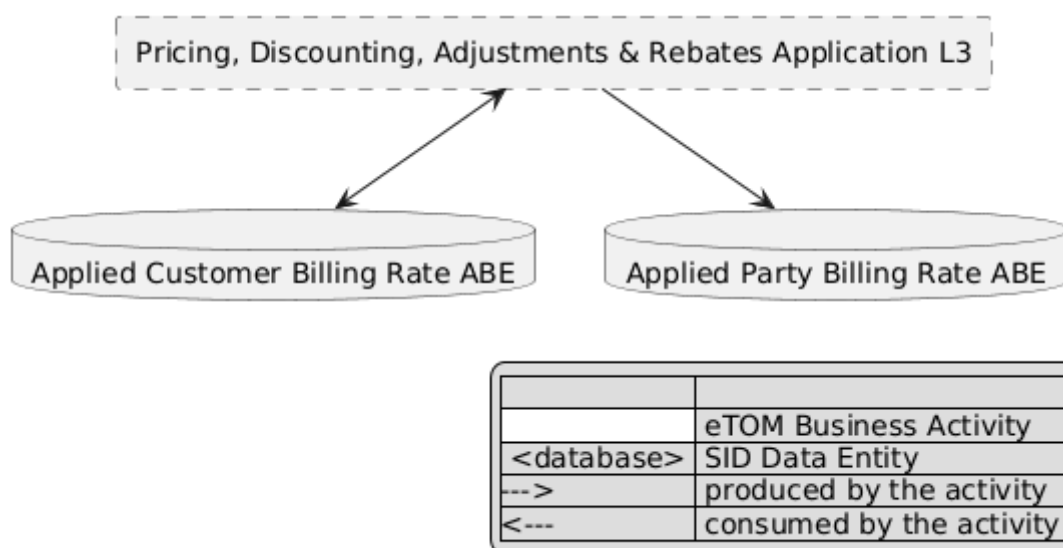
TMFC031 – Bill Calculation

1. Overview

Component Name	ID	Description	ODA Function Block
Bill Calculation	TMFC031	The Bill Calculation component processes all charges against billing accounts during bill cycles. Bill Calculation can be executed both on a cyclic basis and on demand. It performs calculations with bill compilation of charges, credits, fees & taxes, including pro rata, at various levels, such as product and/or account level that have been generated since the last run for that account, applying promotions and discounts.	Core Commerce

2. eTOM Processes, SID Data Entities and Functional Framework Functions

2.1. eTOM L2 - SID ABEs links



2.2. eTOM business activities

eTOM business activities this ODA Component is responsible for:

Identifier	Level	Business Activity Name	Description
1.3.9	L2	Customer Bill Invoice Management	Ensure the bill invoice is created, physically and/or electronically produced and distributed to customers, and that the appropriate taxes, discounts, adjustments, rebates and credits for the products and services delivered to customers have been applied.
1.3.9.4	L3	Pricing, Discounting, Adjustments & Rebates Application	Ensure that the bill invoice is reflective of all the commercially agreed billable events and any bill invoice adjustments agreed between a Service Provider and the customer.
1.3.9.4.1	L4	Obtain Billing Events	Accept billing events that have been collected, translated, correlated, assembled, guided and service rated before determining the information would be applied to the customer's bill invoice(s).
1.3.9.4.2	L4	Apply Pricing, Discounting, Adjustments & Rebates to Customer Account	Determine the customer account or customer specific pricing, charges, discounts, and taxation that should be delivered to the invoice(s) for the customer.
1.3.9.4.3	L4	Apply Agreed Customer Bill Adjustment	Apply and review any adjustment agreed in the previous billing period and make these included to the invoice.
1.6.15	L2	BP Bill/Invoice Management	Business Partner Bill/Invoice Management manages the Business Partner bill/invoice process, controls bills/invoices, manages the lifecycle of bills/invoices. A bill is a notice for payment which is supposed to be preceded by an invoice in most cases.

1.6.15.1	L3	BP Bill/Invoice Process Management	Make certain that there is capability so that the Bill Invoice Management processes can operate effectively and design and develop an enterprise's invoicing process.
1.6.15.3	L3	BP Bill/Invoice Lifecycle Management	Ensure bills/invoices are created, physically and/or electronically produced and distributed to parties, and that the appropriate taxes, discounts, adjustments, rebates and credits for the products delivered to parties have been applied.
1.6.15.3.1	L4	Apply BP Pricing, Discounting, Adjustments & Rebates	Ensure that a bill/invoice is reflective of all the commercially agreed billable events and any bill/invoice adjustments agreed between an enterprise and a BP.
1.6.15.3.5	L4	Receive BP Bill/Invoice	Receive and record the bill/invoice from a BP. Compare a BP bill/invoice against all transactions with the BP that would result in a bill/invoice being sent to the enterprise. Manage the interactions between a BP and an enterprise. Approve a BP bill/invoice.

2.3. SID ABEs

SID ABEs this ODA Component is responsible for:

SID ABE L1	SID ABE L1 Definition	SID ABE L2 (or set of BEs)	SID ABE L2 Definition
Applied Customer Billing Rate	The Applied Customer Billing Rate ABE deals with the correlation of related usage for subsequent rating, rates applied to the usage (both regulated and non-regulated), discounts to usage, and any taxes due on the rated usage.		<i>(no description available)</i>
Applied Party Billing Rate	This ABE is still preliminary		<i>(no description available)</i>

2.4. Functional Framework Functions

Function ID	Function Name	Aggregate Function Level 1	Aggregate Function Level 2	Function Description
67	Usage Summary and Details Presentation	Invoice Management	Invoicing	Usage Summary and Details Presentation present usage summary and details (billed and non-billed) for a specific time period.
87	Billing Event Processing Guiding	Invoice Management	Invoicing	Billing Event Processing Guiding support for a consistent processing.
316	Billing Administration	Invoice Management	Invoicing	The Billing Administration function manages the data that are necessary to perform the bill calculation: billing cycle data, management of runs, groups and cycles of invoicing.
399	Billing Management Integration	Invoice Management	Invoicing	Billing Management Integration provide a Virtual Network Operators online access function to make them self sufficient for Billing management. Including the use of VNO/Dealer data fencing.
32	Billing Initialization	Invoice Management	Invoicing	Billing Initialization initializes the bill and sends to the Bill Calculation application the required information for accounts that are going to be processed.
72	Billing Account Price Plan Determining	Invoice Management	Billing Account Administration	Billing Account Price Plan Determining associates a charge record with the appropriate price plan.

70	Charge To Billing Account Distribution	Invoice Management	Billing Account Administration	Charge To Billing Account Distribution identifies the related prepaid or postpaid billing account for a given charge (recurring, one time, usage).
68	Charges to Billing Statement Identification	Invoice Management	Billing Account Administration	Charges to Billing Statement Identification identifies what charges are to be included in the statement.
158	Commitment Tracking Result Determining	Rating and Follow up	Bill Calculation	Commitment Tracking Result Determining determines the outcome of the evaluation (financial benefits or penalties) in the context of the gathered data for commitment tracking.
159	Commitment Tracking Terms & Conditions Evaluation	Rating and Follow up	Bill Calculation	Commitment Tracking Terms & Conditions Evaluation evaluates the terms and conditions in the context of the gathered data for commitment tracking.
160	Commitment Tracking Data Collection	Rating and Follow up	Bill Calculation	Commitment Tracking Data Collection collects data to be used in the evaluation of the terms and conditions to monitor financial commitments between the customer and the provider.

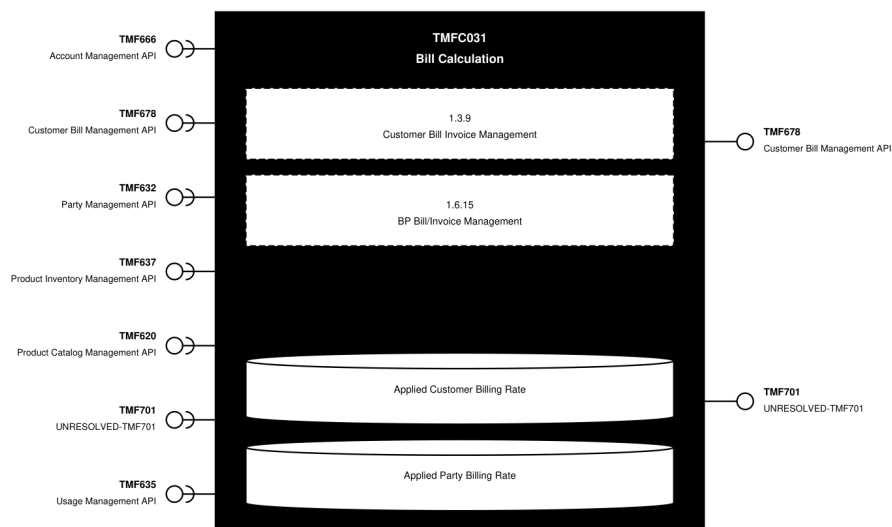
256	Customer Bill Usage and Charges Viewing	Rating and Follow up	Bill Calculation	Customer Bill Usage and Charges Viewing provides an internet technology driven interface to the customer to undertake Usage and charges comparison and unbilled charges view directly for themselves.
89	Billing Event Aggregation	Invoice Management	Invoicing	Billing Event Aggregation is part of the Billing Event Processing to supply aggregated billing events to the Billing System.
90	Billing Event Processing Analyzing	Invoice Management	Invoicing	Billing Event Processing Analyzing provides billing event analysis and billing event aggregations analysis to control the usage data sent to the Billing System.
183	Bill Charges Aggregation	Rating and Follow up	Bill Calculation	Bill Charges Aggregation function determines charges (including recurring, one time and usage charges) for purchased products and services in a given bill run based on the customer price plan set at time of the customer order/contract negotiation.
60	Split Bill Charge Distribution	Rating and Follow up	Tariff Calculation and Rating	Split Bill Charge Distribution provides charge and event distribution to support a split bill.

184	Currency Conversion	Rating and Follow up	Bill Calculation	Currency Conversion identifies the required currency conversion if any needed to appropriately bill the customer.
61	On Demand Bill Calculation	Rating and Follow up	Bill Calculation	On Demand Bill Calculation function will invoke a bill calculation on demand for e.g. a purchase.
300	Discount Calculation	Rating and Follow up	Tariff Calculation and Rating	Discounts Calculation determines charge discounts based on pricing plan; including discounts on recurring, one time, and usage charges. Discounts may be applied at different levels such as cross product, cross location, or cross customer (all customers that are part of a given group plan – some affiliation). The discounts can be apportioned across multiple events.
55	Price and Discount Calculation	Rating and Follow up	Tariff Calculation and Rating	Price and Discount Calculation applies pricing and discounting rules and algorithms in the context of the assembled information concerning Products (i.e. instances of Product).
69	Charge To Billing Account Identification	Invoice Management	Billing Account Administration	Charge To Billing Account Identification associates incurred charge to the billing account liable for its payment.

3. TM Forum Open APIs & Events

The following part covers the APIs and Events; this part is split in 3: - List of Exposed APIs - This is the list of APIs available from this component. - List of Dependent APIs - In order to satisfy the provided API, the component could require the usage of this set of required APIs. - List of Events (generated & consumed) - The events which the component may generate are listed in this section along with a list of the events which it may consume. Since there is a possibility of multiple sources and receivers for each defined event.

3.1. API Context Diagram

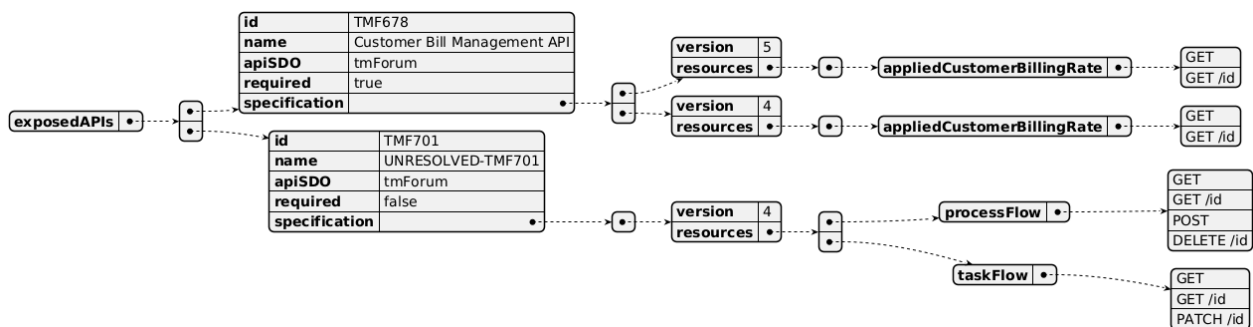


(SVG source: [TMFC031_API_Context.svg](#) — hand-drawn rather than PlantUML for this one diagram, since PlantUML's automatic layout couldn't give each API its own straight, individually-anchored connector at a chosen height. Generated by the *component-specification-documentation skill's scripts/render_api_context_svg.py*.)

3.2. Exposed APIs

API ID	API Name	Mandatory / Optional	API Version	Resource	Operations
TMF678	Customer Bill Management API	Mandatory	5	appliedCustomerBillingRate	GET /id

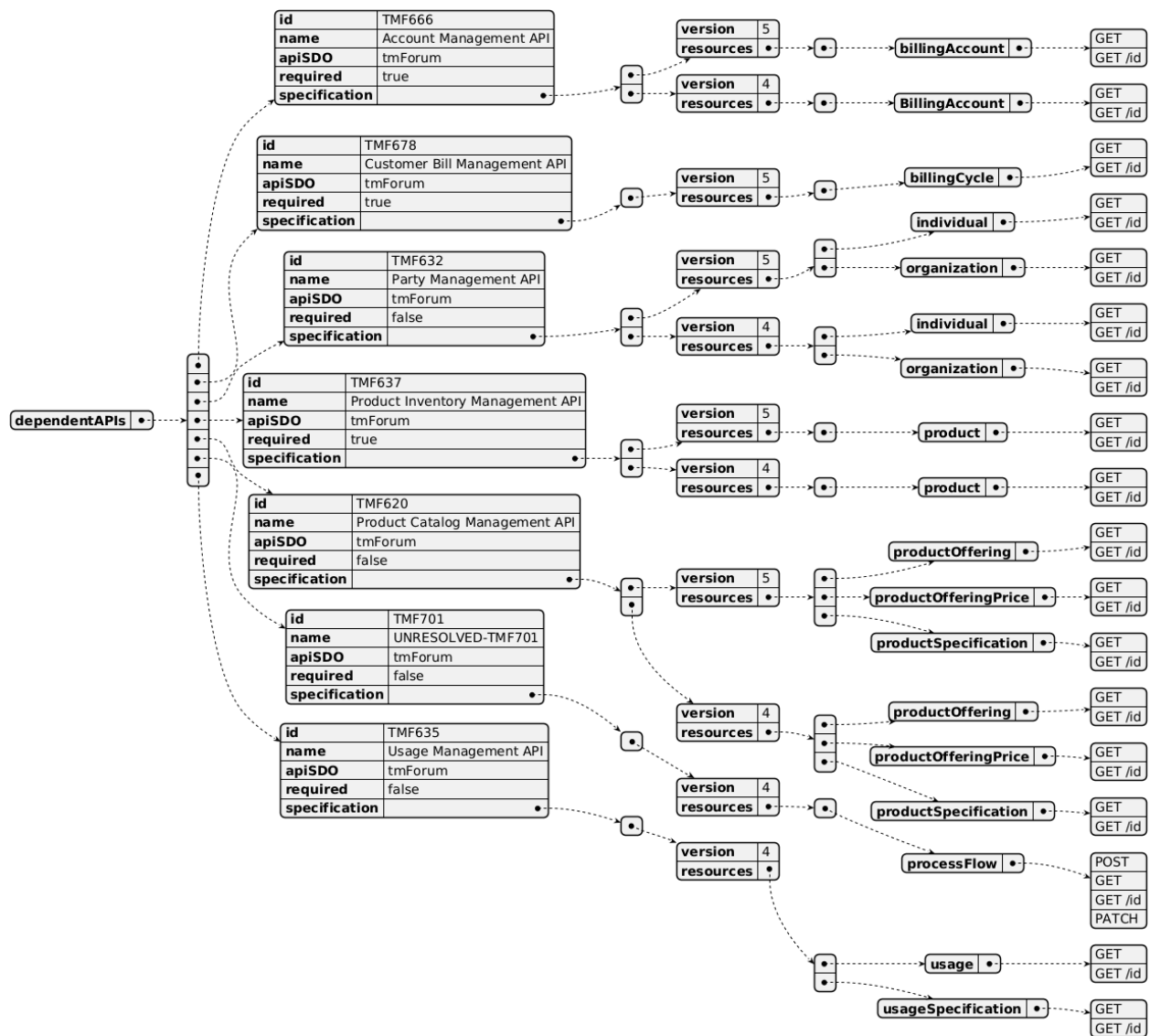
TMF678	Customer Bill Management API	Mandatory	4	appliedCustomerBillingRate	GET, GET /id
TMF701	UNRESOLVED-TMF701	Optional	4	processFlow	GET, GET /id, POST, DELETE /id
TMF701	UNRESOLVED-TMF701	Optional	4	taskFlow	GET, GET /id, PATCH /id



3.3. Dependent APIs

API ID	API Name	API Version	Mandatory / Optional	Resource	Operations
TMF666	Account Management API	5	Mandatory	billingAccount	GET, GET /id
TMF666	Account Management API	4	Mandatory	BillingAccount	GET, GET /id
TMF678	Customer Bill Management API	5	Mandatory	billingCycle	GET, GET /id
TMF632	Party Management API	5	Optional	individual	GET, GET /id
TMF632	Party Management API	5	Optional	organization	GET, GET /id
TMF632	Party Management API	4	Optional	individual	GET, GET /id
TMF632	Party Management API	4	Optional	organization	GET, GET /id
TMF637	Product Inventory Management API	5	Mandatory	product	GET, GET /id

TMF637	Product Inventory Management API	4	Mandatory	product	GET, GET /id
TMF620	Product Catalog Management API	5	Optional	productOffering	GET, GET /id
TMF620	Product Catalog Management API	5	Optional	productOfferingPrice	GET, GET /id
TMF620	Product Catalog Management API	5	Optional	productSpecification	GET, GET /id
TMF620	Product Catalog Management API	4	Optional	productOffering	GET, GET /id
TMF620	Product Catalog Management API	4	Optional	productOfferingPrice	GET, GET /id
TMF620	Product Catalog Management API	4	Optional	productSpecification	GET, GET /id
TMF701	UNRESOLVED-TMF701	4	Optional	processFlow	POST, GET, GET /id, PATCH
TMF635	Usage Management API	4	Optional	usage	GET, GET /id
TMF635	Usage Management API	4	Optional	usageSpecification	GET, GET /id



3.4. Events

The diagram illustrates the Events which the component may publish and the Events that the component may subscribe to and then may receive. Both lists are derived from the APIs listed in the preceding sections.

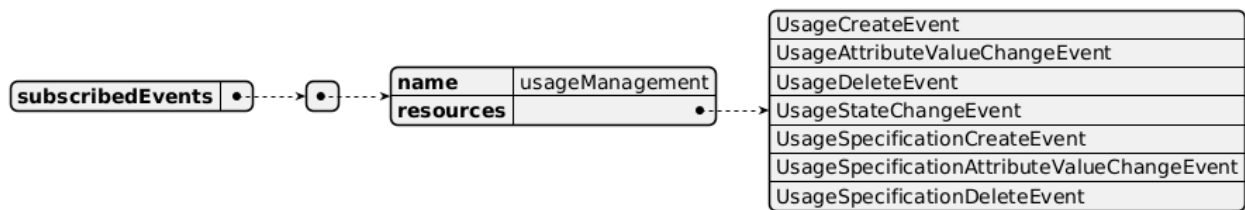
Published Events

(none listed in the component YAML)



Subscribed Events

API ID	API Name	Event Resources
TMF635	Usage Management API	UsageCreateEvent UsageAttributeValueChangeEvent UsageDeleteEvent UsageStateChangeEvent UsageSpecificationCreateEvent UsageSpecificationAttributeValueChangeEvent UsageSpecificationDeleteEvent



4. Machine Readable Component Specification

Refer to the ODA Component Directory on the TM Forum website for the machine-readable component specification files for this component.

5. References

5.1. TMF Standards related versions

Standard	Version(s)
eTOM	v23.5
SID	v23.5.0
Functional Framework	v23.5.0.1

5.2. Jira References

SID - ISA-894 - Formalize association between ProductUsage and AppliedCustomerBillingProductUsageRate (Done): in SID, rated ProductUsage represents AppliedCustomerBillingProductUsageRate. See ticket for more details.

eTOM n/a

Functional Framework - ISA-1032 - Clarification of Bill-time fee and what types it refers to in function 58 (BACKLOG): clarification of Function 58 to understand the types of fees that are underpinned by this function - bill-time fee. - ISA-1033 - FF 73 - what type of reporting does this function cover? (BACKLOG): clarification of Function 73 to understand what type of reporting is meant by the function to determine the component to map to. - ISA-996 - Master Data Management has repurposed catalog related functions (BACKLOG): Function 6 re-purposed by Data Repository Management.

OpenAPI - AP-4740 - Formalize association between ProductUsage and AppliedCustomerBillingProductUsageRate (OPEN) - AP-5471 - TMF767 - updates for consistency with other 2 Usage APIs: 767 Product Usage (DONE): the future of Usage Management APIs moving to Product, Service & Resource from version 5. - AP-6883 - TMF678 - add POST, PATCH, DELETE operations on appliedCustomerBillingRate (BACKLOG): currently only GET operations are available for Applied Customer Billing Rate; this is a request to add POST and PATCH.

ODA Component n/a

5.3. Further resources

1. IG1228: please refer to IG1228 for defined use cases with ODA components interactions.

6. Administrative Appendix

6.1. Document History

6.1.1. Version History

Version Number	Date	Modified by	Description of changes
1.0.0	27-Sep-2023	Gaetano Biancardi	First version of document
1.0.0	19-Dec-2023	Amaia White	Final administrative edits

1.1.0	29-Jan-2024	Elisabeth Andersson	Functional Framework added: 6 Unbilled Invoice Items Listing, 68 Charges to Billing Statement Identification, 69 Charge To Billing Account Identification, 70 Charge To Billing Account Distribution, 158 Commitment Tracking Result Determining, 159 Commitment Tracking Terms & Conditions Evaluation, 160 Commitment Tracking Data Collection, 256 Customer Bill Usage and Charges Viewing
1.1.0	01-Mar-2024	Amaia White	Final administrative edits prior to release
2.0.0	29-May-2024	Elisabeth Andersson, Gaetano Biancardi	Updated to the latest template; added TMF620 Product Catalog as dependent API; modified API optionality (Optional to Mandatory: TMF637, TMF669, 635; removed TMF672 and TMF688 as they will be part of Canvas Services; removed TMF669; removed the need to access billingFormat and billingPresentationMedia from TMF666; updated TMF666 billingAccount and billingCycleSpecification to only have GET and GET/id); updated to 23.5 version of frameworks; eTOM updates: removed 1.3.9.3, added 1.6.15.3.1, added higher-level descriptions for already-included lower levels for 1.6.15; FF updates: changed aggregate action levels for 158, 159, 160, 258, added 60, 61, 89, 90, 183, 184
2.0.0	06-Sep-2024	Amaia White	Final administrative edits prior to publication

3.0.0	27-Oct-2025	Elisabeth Andersson	Removed dependency on TMF666 for the Bill Cycle Specification and instead added TMF678 Bill Cycle operations, providing the instantiated Bill Cycle for a given period required by Bill Calculation; updated ISA-894 as resolved (in SID, rated ProductUsage represents AppliedCustomerBillingProductUsageRate); added Jira ticket dependency to TMF678 - add POST, PATCH, DELETE operations on appliedCustomerBillingRate (BACKLOG)
3.0.0	19-Nov-2025	Rosie Wilson	Updated description; final administrative edits prior to publication
3.1.0	25-May-2026	Hugo Vaughan	Gen 5 API updates
3.1.0	25-Jun-2026	Sannah Sibaya	Final edits prior to publication

6.1.2. Release History

Release Status	Date Modified	Modified by	Description of changes
Pre-production	19-Dec-2023	Amaia White	Initial release
Pre-production	22-Jan-2024	Adrienne Walcott	Updated to Member Evaluated status
Pre-production	01-Mar-2024	Amaia White	Release of v1.1.0
Production	26-Apr-2024	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	06-Sep-2024	Amaia White	Release of v2.0.0
Pre-production	19-Nov-2025	Rosie Wilson	Release of v3.0.0
Production	16-Jan-2026	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	25-Jun-2026	Sannah Sibaya	Release of v3.1.0

6.2. Acknowledgements

This document was prepared by the members of the TM Forum ODA Components & Canvas team.

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